TexCycle AI: An AI-Powered Textile Waste Marketplace and Circular Economy Intelligence Platform for Sri Lanka

COMMON INTEGRATED SOLUTION REPORT

Team member	Student ID	Individual component
Sandeepa S.A.C.	IT22281814	Fabric Intelligence
Siriwardhana N.S.T.N.	IT22215338	Sustainability Intelligence
Madushanka H.M.T.G.	IT22221278	Seller Marketplace and Commercial inrelligence
Thalagala T.A.D.V	IT22281678	Marketplace Predictive Analytics

Report submitted in partial fulfilment of the requirements for the

BSc (Hons) in Information Technology

Specializing in Information

Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure. We have disclosed all use of generative AI and other automated tools in accordance with the module policy. We remain responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this report.

Student name	Student ID	Signature	Date
[Student 1]	[ID]		
[Student 2]	[ID]		
[Student 3]	[ID]		
[Student 4]	[ID]		

Supervisor certification. The above candidates carried out this research project under my supervision.

Role and name	Signature	Date
Supervisor: [Name]		
Co-supervisor: [Name]		

ABSTRACT

Textile waste poses an environmental and resource-efficiency challenge where reusable/recyclable material is difficult to identify, assess, value, and connect to suitable downstream users. This report presents Tex Cycle AI, an integrated digital marketplace and circular-economy intelligence prototype for the Sri Lankan textile-waste context. The proposed solution combines four coordinated components: Fabric Intelligence (image-based fabric, classification and condition assessment), Sustainability Intelligence (circularity, and environmental indicator estimation), Commercial Intelligence (reference pricing, buyer recommendation, and traceability), and Marketplace Predictive Analytics (demand, listing-success, sales-time, and market-trend predictions). The report's development approach follows an applied design-science process involving requirements synthesis, data preparation, model development, API integration, user-interface integration, and component-level evaluation. The project documentation reports prototype results including 89.68% fabric-classification accuracy, 90.87% defect-classification accuracy, price prediction R^2 of 0.9590, buyer-matching accuracy of 83.56%, demand classification accuracy of 99.54%, listing-success accuracy of 83.65%, market-trend accuracy of 98.72%, and sales-time R^2 of 0.7610. These figures are reported as component-level development results, not comparable across tasks. The marketplace dataset is described as 100,000 synthetic records with 36 variables; hence, high scores do not establish real-world performance in Sri Lanka. The integrated contribution is a structured workflow linking material identification, sustainability, commercial matching, marketplace forecasts, and traceable exchange. Further validation using representative real transactions, documented user studies, and operational monitoring is required before production deployment.

Keywords: textile waste, circular economy, artificial intelligence, digital marketplace, predictive analytics

ACKNOWLEDGEMENTS

The team would like to extend their sincere gratitude to the academic staff and research supervisors of the Sri Lanka Institute of Information Technology for their guidance, feedback, and academic support throughout the research and development of Tex Cycle AI.

The team also wishes to acknowledge the collaboration of all four project members in defining the integrated problem, developing the individual intelligence components, discussing shared interfaces, and bringing the component outputs together into a common marketplace workflow. Each member's contribution is documented separately in this report and the relevant individual thesis.

Finally, the team acknowledges the researchers, standards bodies, open-source communities, and technical documentation authors whose publications and tools supported the design, implementation, evaluation, and documentation of the prototype. No external funding or formal stakeholder access is claimed in this report unless separately confirmed in the project records.

TABLE OF CONTENTS

1 INTRODUCTION AND PROJECT CONTEXT 1

1.1 Background and context 1

1.2 Stakeholders and user needs 2

1.3 Alignment with the Sustainable Development Goals 3

1.4 Purpose and scope of the common report 3

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS 5

2.1 Overall problem definition 5

2.2 Root causes and overall research gap 5

2.3 Subproblem and component mapping 6

2.4 Overall aim and group objectives 8

- GO2: Marry fabric and condition data to sustainability and commercial outputs through documented interfaces and shared identifiers. 8
- GO3: Deliver marketplace level predictive outputs on demand, listing success, sales time, and trend, with interpretability around uncertainty. 8
- GO4: Assess component outputs through the lens of task appropriate metrics while documenting provenance of datasets, split strategy, limitations, and risks. .. 8
- GO5: Assess end-to-end integration, validation, exception handling, and user facing interpretability of AI driven recommendations. 8

2.5 Scope assumptions and constraints 8

3 OVERALL SOLUTION STRUCTURE 10

3.1 Integrated solution overview 10

3.2 High level architecture 11

3.4 Component 2 summary — Sustainability Intelligence 12

3.5 Component 3 summary — Commercial Intelligence 13

3.6 Component 4 summary — Marketplace Platform and Predictive Analytics ... 13

3.7 End to end user and data flow 14
4.1 Integration strategy 15
4.2 Interfaces data contracts and shared models 15
4.3 Shared nonfunctional requirements 16
4.4 Integrated workflow and exception handling 16
4.5 Team contribution and dependency matrix 17
5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION 20
5.1 Common technology stack and infrastructure 20
5.2 Development and integration process 21
5.3 Deployment and operational configuration 22
5.4 Evidence of team execution 23
6 INTEGRATED TESTING EVALUATION AND VALIDATION 24
6.1 Evaluation framework and success criteria 24
6.2 Integration and end to end testing 24
6.3 Overall results 25
The provided common-report template asks for stakeholder and user validation. The supporting materials available for this document did not include a verified list of participants, a usability protocol, a consent record, or the final user-study set of data. As such, no claims can be made that formal user acceptance testing or stakeholder validation has taken place. 28
In order to finish this evaluation correctly, the team should conduct a structured walkthrough with a few representative suppliers and buyers, define success criteria for tasks ahead of time, record completion time and errors, gather a short usability questionnaire, and analyze qualitative feedback. It is important to tell the participants that the platform is a research prototype and that its predictions can be wrong. Results should identify sample size and recruitment method and should avoid generalizing from the small convenience sample to the whole Sri Lankan textile sector. 28

6.4 Stakeholder and user validation 28

The provided common-report template asks for stakeholder and user validation. The supporting materials available for this document did not include a verified list of participants, a usability protocol, a consent record, or the final user-study set of data. As such, no claims can be made that formal user acceptance testing or stakeholder validation has taken place. 28

In order to finish this evaluation correctly, the team should conduct a structured walkthrough with a few representative suppliers and buyers, define success criteria for tasks ahead of time, record completion time and errors, gather a short usability questionnaire, and analyse qualitative feedback. It is important to tell the participants that the platform is a research prototype and that its predictions can be wrong. Results should identify sample size and recruitment method and should avoid generalising from the small convenience sample to the whole Sri Lankan textile sector. 28

6.5 Achievement of common objectives 29

6.6 Integrated limitations and threats 30

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION 31

7.1 Value proposition and beneficiaries 31

7.2 Market alternatives and differentiation 31

7.3 Business model and feasibility 32

7.4 Adoption intellectual property and compliance 33

7.5 Sustainable Development Goal contribution 34

8 CONCLUSION AND FUTURE DIRECTIONS 35

8.1 Overall conclusion 35

8.2 Principal integrated contribution 35

8.3 Future directions 36

REFERENCES 38

APPENDICES 39
Appendix A Integrated interface specification 39
Appendix B Team contribution declaration 40
Appendix C — Integrated Test Matrix 40
Appendix D — Model and Data Provenance Checklist 41
Appendix E — Team Contribution Declaration 42
Appendix G — Risk Register 44
Appendix H — Recommended Release Checklist 45
Appendix M — Sustainability Impact Measurement Plan 57
TEMPLATE CONVENTIONS 61

LIST OF FIGURES

[Guidance - delete before submission: Use Word's Insert Caption command with the label Figure. Number figures by chapter and cite every figure in the text.]

Right-click and select Update Field

LIST OF TABLES

[Guidance - delete before submission: Use Word's Insert Caption command with the label Table. Number tables by chapter and cite every table in the text.]

Right-click and select Update Field

LIST OF ABBREVIATIONS

[Guidance - delete before submission: List abbreviations alphabetically. Define each abbreviation at first use in the main text even when it appears here.]

Abbreviation	Meaning
[ABC]	[Full term]
[ABC]	[Full term]
[ABC]	[Full term]

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Textile production and consumption generate a stream of offcuts, deadstock, rejected fabric, post-industrial remnants, and post-consumer textiles. Some of these materials are valuable for direct reuse, repair, remanufacturing, insulation, cleaning products, accessories, or fiber recovery – but value is not easily realized: a potential buyer needs to know what a material is, its condition, available quantity, location, and whether it is suitable for a particular use. When this information is fragmented or inconsistent, suppliers may struggle to find buyers and buyers may struggle to compare materials.

The textile-waste challenge is therefore not just a disposal problem – it is an information, coordination and trust problem. Circular-economy approaches seek to keep products and materials in use for as long as possible and reduce avoidable waste, but in practice such strategies require reliable material information, viable exchange pathways, and transparent decision-making [1], [2]. Digital systems can help in this process by structuring descriptions, assisting classification, communicating likely value, and connecting supply with demand. Such systems should complement – rather than replace – material inspection, business judgement, and relevant environmental evaluation.

Sri Lanka's textile and apparel sector is economically important and participates in international supply chains. The proposed project relates to a digital marketplace that could help textile suppliers and downstream users learn about potential reuse and recovery opportunities. This report does not claim that every waste stream could be recycled, or that every transaction will produce a quantified environmental benefit. These will depend on material composition, processing requirements, transport, substitution effects, and the end use.

1.2 Stakeholders and user needs

The principal stakeholder groups are textile manufacturers and suppliers, small and medium-sized upcycling businesses, designers and artisans, recyclers, manufacturers using recovered textile material, marketplace administrators, and researchers evaluating the platform. Each group has disparate information needs and control needs.

Suppliers need a streamlined description process, recommendation reviewing, price setting, publishing, and tracking. Buyers need accessible data search, quantitative and qualitative comparisons, realistic quantity estimates, material conditions, and seller contact or proposal options. Administrators need moderation tools, auditing, and invalid or misleading record management. The research team needs model evaluation reproducibility, and differentiation between prototypical predictions.

The stakeholder needs should be rephrased as design requirements, stemming from the project scope. One should refrain from labeling requirements as established if the team used interviews or user studies for research, unless there are questionnaires, consents, transcripts, or test results. The requirements should be justified by a subsequent validation phase, which will document the selection process, questions, participants, consents, results, and design changes.

- Material clarity: consistent textile type, condition, quantity, and location fields.
- Decision making interpretable reference price, likely buyer categories, and marketplace forecasts.
- Trust: visible limitations, data provenance, model version, and traceable listing events.
- Operational control: role-based access, validation, moderation, and recoverable error handling.

1.3 Alignment with the Sustainable Development Goals

Tex Cycle AI aligns itself with Sustainable Development Goal 12 (Responsible Consumption and Production) in terms of its primary mission, which is consistent with the goals of 'reducing consumption, promoting waste prevention and reduction, encouraging recycling and re-use'. The platform also contributes to Sustainable Development Goal 8 (Decent Work and Economic Growth) if it manages to identify potential flows of recyclable material and create economic value. In turn, its contribution to the United Nations Sustainable Development Goals (UNSDG) does not necessarily mean a significant reduction of its impact on a particular SDSG target [3].

Any claims of the software's impact should be based on physical material flow information and supported by an analytical framework. For instance, a software-based estimate of kilograms of material diverted from landfill disposal does not equal a reduction of landfill use if the location of this material and the alternative it would have represented were not identified. Similarly, a prediction of carbon or water saved requires a clear description of system boundaries, reference systems, emission factors, and uncertainty analysis [4], [5]. The solution therefore tends to present sustainability performance indicators as decision-support tools and approximations of potential impact rather than independent certifications of their environmental performance.

1.4 Purpose and scope of the common report

This common report contains the shared problem description, integrated solution architecture, responsibilities of the four components, data and API dependencies, implementation approach, testing strategy, evidence of prototype, commercial feasibility and limitations. This report is different from the four individual theses – those reports contain the detailed model methods, component-specific implementation, experiments, personal contributions and their own evidence.

The structure and formatting follow the supplied SLIIT documentation guidance [19], while the project specific component boundaries and evidence are taken from the team's internal documentation [20], [21].

The report covers the integrated prototype and the evidence that was provided in the project documentation. It does not claim about production readiness, audited security, verified real-world market performance, a deployed payment gateway, or a completed independent user study, unless they are evidenced. The report uses terms of logical design or recommended practice, where the implementation details are not confirmed.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The common problem addressed by the project is that textile-waste suppliers and potential downstream users do not have a reliable, consistent, and decision-supportive digital process for identifying materials, determining reuse potential, and assessing commercial value for matching to appropriate buyers.

The problem statement is deliberately distinguished from the proposed technology: the issue arises when information and coordination barriers prevent viable textile streams from reaching suitable users; AI and a marketplace are possible responses (not the problem itself). A successful system must demonstrate that its outputs are useful, comprehensible, and incorporated into an exchange workflow, not simply that models can produce predictions.

2.2 Root causes and overall research gap

Several related factors motivate my project. First, textile waste has a heterogeneous nature – it is different in terms of fabric type, blend, condition, defect level, and remaining usable life from one lot to another. Second, commercial characteristics are often incomplete – price depends on amount, location, quality, demand, competition, buyer requirements, and transaction context. Third, environmental indicators vary depending on the system boundaries and assumptions and should not be considered universal constants. Fourth, markets need trade-enabling infrastructure such as identity, listing moderation, offer handling, delivery status, and dispute resolution.

An isolated solution can address one aspect but will leave the overall business process fragmented. An image classifier alone cannot identify potential buyers; a sustainability score alone cannot enable a sale; a price estimate alone cannot confirm that a person will buy; a marketplace listing without forecasting cannot help plan. The research gap I want to address is designing and evaluating a coordinated workflow that uses material intelligence, sustainability interpreters, commercial matchers, and marketplace performance estimators while reflecting the uncertainty and evidence limitations of these workflows.

planning support. The integrated research gap is therefore the design and evaluation of a coordinated workflow that connects material intelligence, sustainability interpretation, commercial matching, and marketplace performance estimates while communicating uncertainty and evidence limitations.

Problem area	Root cause / challenge	Integrated response	Evidence needed
Material identity and condition	Variable fabrics and inconsistent descriptions	Fabric classification and condition assessment	Held-out image test set; class-level metrics; failure review
Environmental interpretation	Impacts depend on material, process and baseline	Sustainability indicators with explicit assumptions	Model metrics plus validated factors and boundary documentation
Commercial exchange	Price and buyer suitability vary by context	Reference price and ranked buyer suggestions	Regression metrics, matching evaluation, user acceptance evidence
Marketplace planning	Demand and sale outcomes are uncertain	Demand, listing-success, sales-time and trend predictions	Temporal/held-out evaluation and leakage checks
Trust and integration	Disconnected records and opaque outputs	Shared identifiers, API contracts, traceability events	Integration tests, audit records, security and usability evidence

Table 1 Common problem areas, root causes and integrated responses

2.3 Subproblem and component mapping

The project splits the common problem into the four parts. Each part has its own task or analytical aspect. The division helps avoid repetition and allows every individual thesis statement to stand out against the background of the common report of the outcomes of all the researchers.

Component	Primary question	Inputs	Outputs / responsibility
Member 01 —	What fabric	Textile image and	Fabric class, confidence,

Fabric Intelligence	type and visible condition are suggested by the submitted image?	supporting listing attributes	defect/condition indicators
Member 02 — Sustainability Intelligence	What circularity and environmental indicators can be estimated from structured material information?	Fabric type, weight, condition, health/repairability, location and industry fields	Circularity/recyclability and estimated carbon/water indicators; sustainability grade
Member 03 — Commercial Intelligence	What reference price and buyer categories may fit this material, and how can its lifecycle be recorded?	Material attributes, quantity, location, condition, market context	Reference price, ranked buyer categories, traceability events
Member 04 — Marketplace Predictive Analytics	How might a listing perform in terms of demand, success, time to sell and market trend?	Material and sustainability attributes plus listing and market-context features	Demand level, sold/unsold probability, estimated days to sell, trend class

Table 2 Subproblem and component mapping

2.4 Overall aim and group objectives

- GO1: Develop a common material and listing workflow which ties the four components together without duplicating component ownership.
- GO2: Marry fabric and condition data to sustainability and commercial outputs through documented interfaces and shared identifiers.
- GO3: Deliver marketplace level predictive outputs on demand, listing success, sales time, and trend, with interpretability around uncertainty.
- GO4: Assess component outputs through the lens of task appropriate metrics while documenting provenance of datasets, split strategy, limitations, and risks.
- GO5: Assess end-to-end integration, validation, exception handling, and user facing interpretability of AI driven recommendations.
- GO6: Evaluate the commercial and sustainability value of the proposed solution without overstating environmental or market outcomes that have not been measured.

2.5 Scope assumptions and constraints

The prototype scope involves developing an information system for sellers and buyers, image recognition material intelligence, sustainability estimates, reference price and buyer advice, marketplace predictions reports, and relevant records. The implementation of the front-end roles and operations that have been prototyped must be validated using the existing repositories and deployments. The marketplace dataset is documented as 100,000 synthetic records with 36 variables. It is useful for development and pipeline experimentation, but it is not a representative sample of actual Sri Lankan marketplace transactions. Synthetic labels might encode some assumptions or regularities that the models are likely to learn and reproduce as patterns, so the best performance may reflect agreement with the constructed labels rather than real-world transactions.

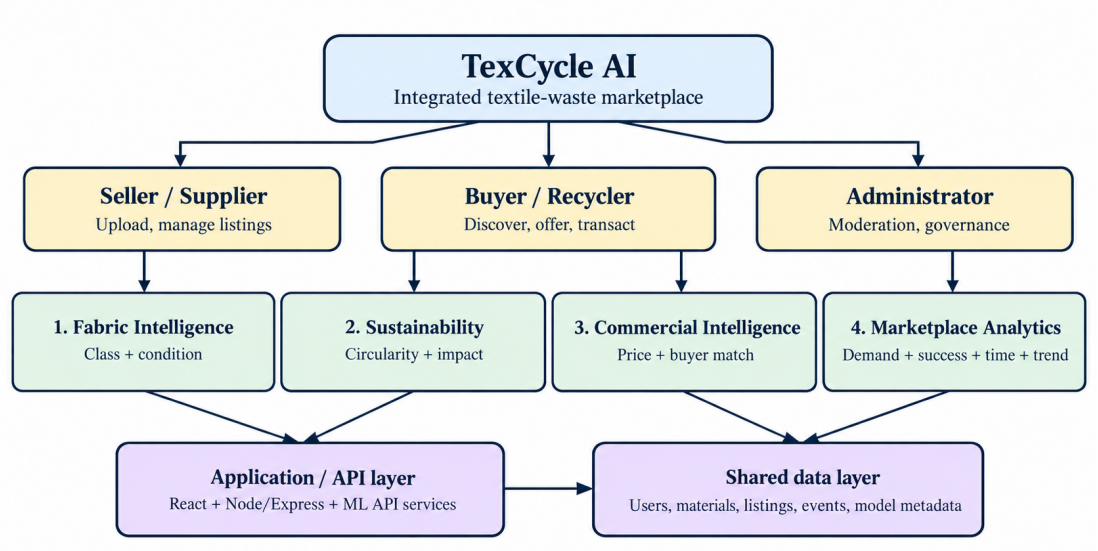
The critical constraints include insufficient live market data, limited transaction records, unreliable seller-buyer feedback, potential input-output mismatches, class imbalance, distribution shifts, and legal data handling issues. The prototype does not claim to make absolute predictions, conduct independent environmental audits, or replace professional textile laboratory assessments.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

Tex Cycle AI is an integrated marketplace and intelligence workflow. Sellers initiate by establishing a textile lot, providing an image and structured attributes. Fabric Intelligence informs material and condition signals. Sustainability Intelligence utilizes relevant structured fields to estimate circularity and environmental indicators. Commercial Intelligence provides a reference price and potential buyer categories. Marketplace Predictive Analytics informs forecasts about demands, listing success, time to sell and trend. The seller may then review and publish the listing, and buyers can discover materials, interact through the marketplace's exchange functions.

The value of the integrated approach is the relationships between the outputs. For example, material type and condition may impact who buyer suitability might be; quantity and location may impact commercial attractiveness; sustainability attributes may assist with filtering and interpretation and observed marketplace outcomes may become training data after they are verified and governed. The system must preserve the provenance of each field, and distinguish user-provided values, computer-vision predictions, model estimated values and confirmed transaction facts.

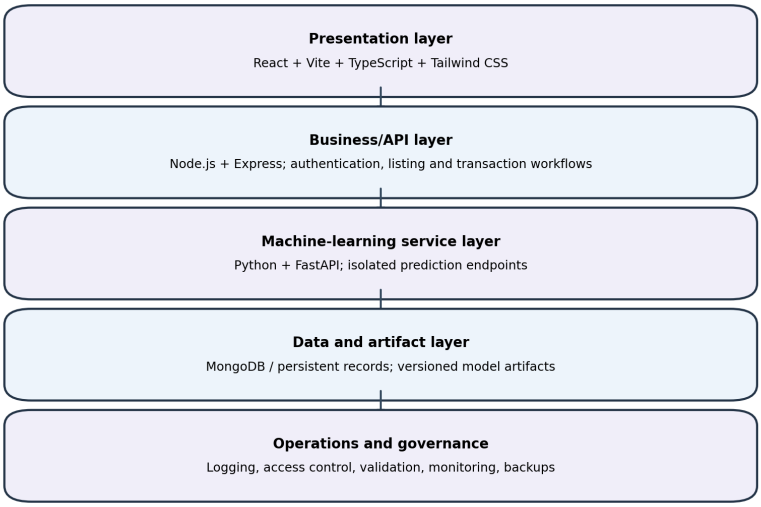


Conceptual architecture based on project documentation; deployment topology must be verified against the actual environment.

3.2 High level architecture

The logical architecture of the project can be split into five layers. The presentation layer includes React interfaces for different users. The business/API layer manages authentication, listings, buyers, offers, and records. The machine-learning service layer provides model inference through separate endpoints, which are described in the project as Python/FastAPI services. The data and artifact layer stores application records and model files separately. Finally, the operations and governance layer manages logging, permissions, monitoring, backup, model versioning, and record reviews.

Decoupling application business processes from model inference procedures allows designing test and update individual parts of the system without disturbing others. Nevertheless, different endpoints for the application and the machine learning model can introduce risks such as schema mismatches, incorrect categorical feature encoding, request timeouts, or fields absence if the application misses any parameters expected by the model. All these issues can be resolved by making contracts between the application and the model, validating input data, versioning model artifacts, and defining default behaviors.



Logical integration view. Exact hosting, database topology, and CI/CD configuration should be confirmed from deployment evidence.

3.3 Component 1 summary — Fabric Intelligence

Fabric Intelligence tackles the problem of material identification and visible condition assessment. The project's documentation references an EfficientNet-B3-based fabric-classifying model that incorporates 16 distinct fabric categories and a MobileNetV2-based binary defect classifier. The achieved test results, as stated in the documentation, amounted to 89.68% fabric-classification accuracy and 90.87% defect-classification accuracy, which need to be evaluated in the context of the database, split, class distribution, image quality, and assessment procedure described in the corresponding thesis. [7], [8].

The output presents downstream components with a structured fabric category and relevant condition-related attributes. Where appropriate, it reports confidence or uncertainty and allows user correction of an incorrect class. The defect pathway does not describe a process of pixel-level segmentation unless the actual model produces and validates pixel masks; the project documentation makes a point of distinguishing between whole-image binary classification and segmentation.

3.4 Component 2 summary — Sustainability Intelligence

A sustainability intelligence model converts structured textile attributes to circularity and environmental indicators. The described pipeline used separate XGBoost regressors for six output variables plus a single sustainability-grade classifier. Its raw inputs were fabric type, weight, health score, repairability, condition, district, province, and industry type, and its derived indicators were energy saved, landfill diverted, SDG impact, and economic impact. [6].

The development results for the six regression outputs are presented in the following table: circularity MAE = 2.429, RMSE = 3.043, R2 = 0.9657; recyclability MAE = 2.459, RMSE = 3.088, R2 = 0.9502; carbon-reduction MAE = 2.397, RMSE = 3.009, R2 = 0.9589; water-reduction MAE = 2.464, RMSE = 3.084, R2 = 0.9389; CO2-saving MAE = 4.456, RMSE = 6.110, R2 = 0.9887; water-saving MAE = 310.365, RMSE = 437.567, R2 = 0.9854. The project states that the development results rely

on synthetic or rule-based data, thus limiting the conclusions about actual environmental impact.

3.5 Component 3 summary — Commercial Intelligence

Commercial Intelligence is about AI-assisted reference pricing, recommendation of buyers, and traceability. The project documentation mentions a price model built with the XGBoost algorithm, with MAE of 12 24, RMSE of 15 40, and R^2 of 0 9590 on the evaluation set, plus 83 56% matching accuracy for buyers. The reference price is a decision support value, not a binding market price or a guarantee of sale. Buyer recommendations are ranked according to how suitable their materials were with respect to the target material and its context.

Traceability connects the material lot with seller upload, AI analysis, marketplace listing, offer, transaction, delivery, and buyer feedback events. The events should be linked with stable identifiers, timestamps, actor references, and status changes. The model version and input lineage should be captured if a prediction has a material impact on listing decisions

3.6 Component 4 summary — Marketplace Platform and Predictive Analytics

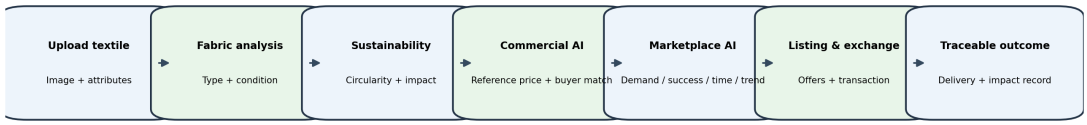
Marketplace Predictive Analytics deals with the possible performance of the marketplace. Project documentation lists an XGBoost classifier for listing success (sold/unsold), an XGBoost regressor for the estimated days to sell, a demand-level classifier, and a market-trend classifier with Growing, Stable, and Declining classes. The listed accuracy is listing-success accuracy 83.65%, sales-time R^2 0.7610, demand accuracy 99.54%, and market-trend classification accuracy 98.72%. These are the results of the development process that should not be confused with live-market performance as described in multiple experiments.The component's practical value is to allow users to evaluate the listing scenarios, manage the inventory, and account for uncertainties. The displayed class probability for a single prediction should not be conflated with the accuracy of the model as a whole, and the user interface should declare the result and, where relevant, the time horizon for the

forecast. The interface should not use language that could mislead the user into believing that an imminent sale is guaranteed or that a certain trend is inevitable.

3.7 End to end user and data flow

The end-to-end process starts with a seller's textile record and ends with the listing outcome and, if completed, traceable transaction events. The data should flow with a shared listing or material identifier. Each step should receive only the data necessary to perform its task and respond with an expected format. The orchestration layer must retain the original user data, model output with model version, and timestamp.

After the listing is posted, the marketplace insights, such as views, wishlist additions, offers, and sales, can provide useful feedback. However, the data should not be used as ground truth since the traffic can be bot-driven, buyers can take items off the wish list, and the status can be changed unintentionally. If the data is to be used for retraining, a versioned dataset with a temporal validation strategy should be created.



Feedback loop: observed views, offers, sales and user feedback can inform future retraining only after quality checks and governance.

The diagram represents the intended integrated workflow, not a claim that every lifecycle event is already automated.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The integration strategy is interface-first and contract driven. Each component should define its required inputs, output fields, error states, and model-version information before being wired into the seller workflow. Validation starts with an isolated model and proceeds to API contract tests, followed by UI integration and end-to-end scenario tests. A component is integrated only when the consuming layer can process valid responses, invalid inputs, timeouts, and unavailable model artifacts.

The components have different update cycles, and versioning is important. A fabric model may change its class vocabulary; a sustainability model may change a unit and a preprocessing pipeline; a price model may change feature requirements; and a marketplace model may require a new target definition. The API should expose or log a model version and schema version so that changes can be traced and regression tested.

4.2 Interfaces data contracts and shared models

Contract area	Recommended fields / rule	Validation requirement
Material identity	material_id, fabric_type, image_reference, condition	Stable ID; allowed class values; image availability
Quantity and location	weight_kg, district, province	Positive quantity; supported location values
Sustainability output	indicator_name, value, unit, model_version, created_at	Units and ranges documented; derived estimates labelled
Price output	predicted_price_per_kg, currency, model_version	Currency/unit explicit; finite non-negative value
Marketplace forecast	demand_level, listing_success_probability, estimated_days_to_sell, market_trend	Probability in [0,1]; days non-negative; class vocabulary fixed
Lifecycle event	event_id, material_id/listing_id, event_type, actor_id, timestamp	Immutable event identity; authorised actor; valid state transition
Error response	error_code, message, request_id, retryable	No sensitive internals; actionable client behaviour

Table 3 Shared data contracts and validation rules

4.3 Shared nonfunctional requirements

Functional correctness is essential but insufficient for a marketplace dealing with business information and AI outputs. The shared requirements which are non-functional are usability, security, reliability, performance, maintainability, accessibility, privacy, auditability and responsible model behaviour. These requirements must be tested across the integrated solution rather than assumed because each component operates independently.

Requirement	Expected behaviour	Evidence / test
Usability	Users understand field labels, units, prediction meaning and next action	Task-based usability test; error and completion observations
Security	Protected operations require authentication and role authorisation	Negative permission tests; dependency and configuration review
Privacy	Collect only required personal/business data and limit access	Data inventory; retention and access review
Reliability	A failed model call does not corrupt the listing record	Timeout, retry and partial-failure tests
Performance	Typical requests complete within a stated target under a defined load	Repeatable load test; report latency percentiles
Maintainability	Model and API versions are traceable; changes have regression tests	Release notes; test results; version metadata
Accessibility	Core workflows support keyboard use, readable contrast and clear errors	Manual accessibility checklist and assistive checks
Explainability	Predictions are described as estimates with limitations	UI review and user comprehension checks

4.4 Integrated workflow and exception handling

The normal process should be resilient to partially or incorrectly provided input. A missing image would still cause image inference to fail, but the user should still be able to save their draft. A model timeout should tell the user that there is no result, instead of inventing a result. In the case of a successful sustainability service call and

a failed marketplace service call, the app should retain the results and allow the user to retry just the failing step if it's technically possible. Duplicate requests should be treated as idempotent operations when they create persistent records or trigger events.

Errors should be classified as validation errors, permission errors, not-found errors, rate-limiting errors, timeouts, model-artifact errors, or unexpected server errors. The UI should display a human-readable summary of the error without low-level implementation details such as stack traces, auth tokens, file system paths, or other implementation details. Server logs should contain request identifiers and relevant diagnostic information for debugging without logging unnecessary or sensitive information.

4.5 Team contribution and dependency matrix

The for the integration. The names/IDs mentioned in the source template areprovided for the identification of the group roles, therefore, the list has to be copied from the approved team list in the corresponding university account upon submission. The matrix below was designed to record the role ownership, not invent the specific individual identities.four-member structure is supposed to offer clear ownership with shared responsibility

Role	**Primary responsibilities**	**Dependencies**	**Shared evidence**
Member 01 — Fabric Intelligence	Fabric classification; condition/defect inference; model and API documentation	Provides fabric/condition features to sustainability, commercial and marketplace layers	Model card, held-out metrics, error cases, interface tests
Member 02 —	Sustainability regressors/classifier;	Consumes fabric and listing	Metric table, preprocessing notes,

Sustainability Intelligence	indicator units and interpretation	attributes; provides indicators to commercial and marketplace layers	API tests, assumptions
Member 03 — Commercial Intelligence	Reference pricing; buyer recommendation; traceability design/events	Consumes material and sustainability attributes; exchanges with listings and transactions	Price metrics, matching evaluation, event schema
Member 04 — Marketplace Analytics	Demand, listing success, sales-time and trend prediction; marketplace integration	Consumes listing/material features and market context; depends on listing status and event feedback	Classification/regression metrics, leakage checks, integration tests
All members	Shared requirements, interface agreement, end-to-end tests, documentation, risk review	Cross-component integration	Meeting records, agreed test matrix, version history, contribution declaration

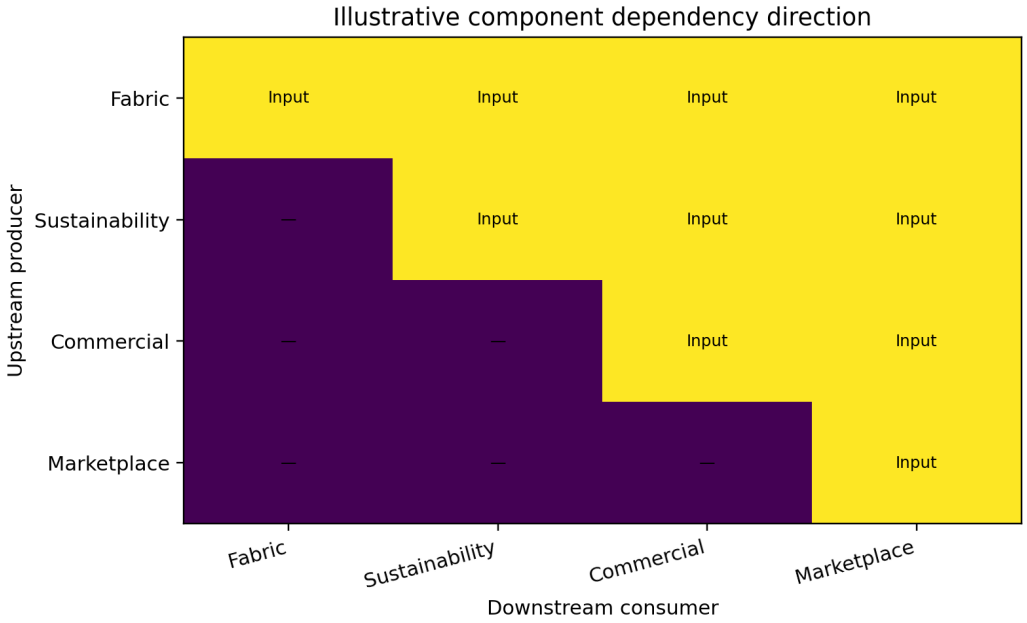


Figure 1 Illustrative component dependency direction. This is a conceptual view; confirm exact calls and dependencies against the current repository.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The project documentation was created for a web application built with React, Vite, TypeScript, Tailwind CSS, React Router, and Lucide icons. The app uses Node.js and Express to power business logic and market APIs and Python and FastAPI to deliver machine learning models' prediction endpoints. Finally, MongoDB and Mongoose are used to support the application's data layer and provide database storage.Overall, the listed tools were used to separate the presentation (UI), business logic, data storage, and machine learning inference parts of the application. [10]–[14].

The following stack summary provides a list of technologies that are supposed to be implemented in the project. There is no indication that all components would be integrated into one production environment and set up with automated CI/CD, monitoring, and backups. The statement needs repository configurations, deployment, infrastructure, or operational information to provide such a conclusion.

Layer	**Documented technology / artefact**	**Purpose in the integrated solution**
Frontend	React, Vite, TypeScript, Tailwind CSS, React Router, Lucide	Role-based UI, listing forms, analysis and forecast presentation
Business backend	Node.js, Express.js	Authentication/authorisation, listing and marketplace operations
Database layer	MongoDB, Mongoose (documented stack)	Persist application entities and related workflow records
ML service	Python, FastAPI	Serve trained models and structured inference responses

Fabric models	EfficientNet-B3 classifier; MobileNetV2 binary defect classifier	Fabric class and image-level condition/defect signals
Sustainability models	XGBoost regressors and sustainability-grade classifier	Environmental/circularity estimates
Commercial models	XGBoost price regressor; buyer recommendation models	Reference price and ranked buyer categories
Marketplace models	XGBoost listing-success classifier; sales-time regressor; demand and trend classifiers	Marketplace decision support
Development support	Version control, issue tracking and technical documentation	Collaboration and reproducibility; exact tools require evidence

5.2 Development and integration process

A proper method for this system's development would be to begin with an analysis of the shared problem and component boundaries. Then, it will be necessary to identify data fields and contracts, develop components, evaluate their outputs, and integrate services into the common seller workflow. It is essential to note that the integration should be incremental to make sure that contract issues are discovered before proceeding to the next stage.

The process of version control should be applied to the source code, model-training scripts, preprocessing pipelines, schemas, and documentation. A record of a particular model release should include information on code version, model artefacts, configuration, dataset version, and test results. Model files should not be updated separately from the preprocessing objects and label encoders that were used in the training process, especially if categorical encoders or class mappings are persisted alongside the model. [16].

Team coordination evidence must be taken from records that have been actually produced, such as: repository history, issue tracker exports, meeting minutes, research diaries, task boards, and reviewed pull requests. This report is not going to make up sprint dates, attendance, commit counts, or task completion statistics. The final team should put in the appendix the evidence of choice after verifying that it is genuine and shows contribution rather than just activity.

5.3 Deployment and operational configuration

The logical deployment model provides for the separation of the web client, business API, database, and machine-learning service. The configuration values, including service URLs, origins, secrets, database credentials, and model paths, should be set for each environment individually, and the secrets and other sensitive information should be stored encrypted. The development, test, and production environments should have different sets of data and credentials.

Prior to deployment, the team should ensure that all required artefacts were provided with the model, check their hashes or versions, set dependencies to specific versions, and test the system's health checks and error responses. It is necessary to store request IDs, durations, error types, and model versions in logs and include that information in the backup. The backup should be validated by attempting to restore it, as verifying the existence of a backup script or job is not sufficient proof that it will work correctly when needed. If the information about the hosting was not provided by the team, this report states the requirements for the deployment environment, not the actual state.

The production-ready review should include a secret management audit, TLS configuration check, database security assessment, dependency vulnerability analysis, rate-limiting and monitoring tool validation, backup retention policy evaluation, incident management procedure review, and rollback mechanism verification.

5.4 Evidence of team execution

The evidence package for the common report should show that the integrated solution was developed collaboratively and that the final version corresponds to the tested version. Appropriate evidence includes a repository overview, commit or pull-request samples, agreed interface specifications, selected issue records, model version notes, and a signed contribution statement. Screenshots should be genuine captures of the actual application or tools and should be dated or otherwise contextualized where needed.

No actual application screenshots or verified project-management exports were included with the two files supplied for this task. Therefore, this report uses original technical diagrams to explain the architecture and workflow, and it does not represent those diagrams as screenshots of a running application. Genuine screenshots should be added in the designated evidence appendix after capture from the current build.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The evaluation framework distinguishes between metrics at the level of models, APIs, end-to-end systems, and stakeholder-based validation. These reflect different aspects of the system, and each is appropriate to distinct sets of questions. A metric for a particular model reflects performance on a given dataset; an API test reflects whether a service responds correctly to particular inputs; an end-to-end test reflects the ability of the system to support a given scenario; and validation by users reflects whether the system is comprehensible and valuable.

Every reported result should specify the task, dataset, split, metric, and provenance. For classification, report accuracy, along with precision, recall, F1-score, and a confusion matrix if feasible. For regression, report MAE and RMSE (in the target's units) and R^2. Predictions on temporal marketplace data should be evaluated using time-aware splits when foreseeable to avoid leakage of future information. Preprocessing steps should be fit only on the training data and applied consistently to validation/test sets. [15]

6.2 Integration and end to end testing

The integrated test plan should include valid and invalid input testing, boundary value analysis, missing fields, timeouts for service availability, model-artifact accessibility, permission checks, duplicate requests, and state transitions. For example, as a seller, I want to create a draft material record and get access to available analysis results and forecasts to publish a listing and see the record’s state. For a buyer, searching and listing details should match material/listing IDs.

The security test plan should include unauthorized access attempts, role boundary tests, input validation, and the absence of sensitive data in response/log messages. For performance testing, the plan should define the load scenario and report latency percentiles instead of focusing on the best response. User acceptance tests should

follow a documented protocol, involve representative users if possible, obtain informed consent, and provide a transparent account of limitations.

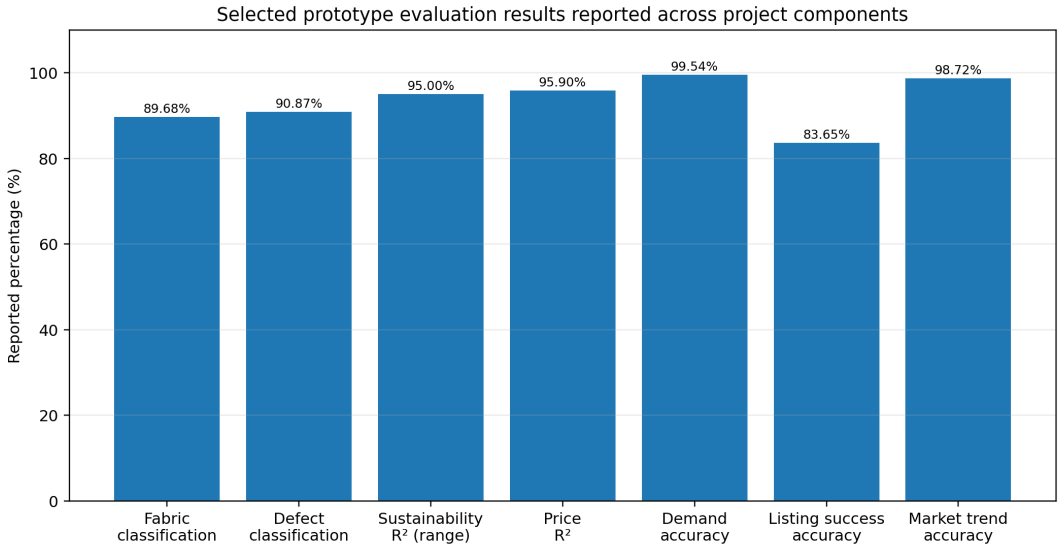
6.3 Overall results

The project source documentation contains the data reported in Table 6.1 describing the component-level metrics. The results reported are not derived from the same data/task and should not be combined to produce an aggregate accuracy metric. The R-squared values reported for the sustainability are regression metrics and should not be interpreted as classification accuracies. The marketplace classification results require more information on class distribution, stratification approaches, and target definitions to be interpreted as effective operational performance.

Component / task	Reported metric	Reported value	Interpretation and caveat
Fabric classification	Accuracy	89.68%	Held-out task metric as reported; dataset composition and per-class performance should accompany it.
Defect classification	Accuracy	90.87%	Binary whole-image classification; not pixel-level segmentation.
Circularity score	MAE / RMSE / R^2	2.429 / 3.043 / 0.9657	Regression result on development data; real-world validity depends on labels and target construction.
Recyclability score	MAE / RMSE / R^2	2.459 / 3.088 / 0.9502	Interpret against target scale and validation data.
Carbon reduction	MAE / RMSE / R^2	2.397 / 3.009 / 0.9589	Environmental estimates require defensible factors and

			boundaries.
Water reduction	MAE / RMSE / R^2	2.464 / 3.084 / 0.9389	Unit and target definition should be stated.
CO_2 saving	MAE / RMSE / R^2	4.456 / 6.110 / 0.9887	Not independently measured savings unless validated against real process data.
Water saving	MAE / RMSE / R^2	310.365 / 437.567 / 0.9854	Target unit and scale should be made explicit.
Price prediction	MAE / RMSE / R^2	12.24 / 15.40 / 0.9590	Price unit/currency and target distribution must be stated in individual thesis.
Buyer matching	Matching accuracy	83.56%	Requires definition of correct match and evaluation sample.
Demand prediction	Accuracy	99.54%	High score on synthetic/derived data may reflect label-generation rules.
Listing success	Accuracy	83.65%	Sold/unsold classification; class balance and leakage checks are important.
Sales time	R^2	0.7610	Regression estimate; report MAE/RMSE and target unit in the detailed thesis.
Market trend	Accuracy	98.72%	Growing/stable/declining class result; temporal validation and class-wise metrics are needed.

Table 4 Reported component-level evaluation results



Note: Values come from different tasks and datasets; the sustainability bar is an illustrative midpoint of reported R^2 values, not an accuracy score. Synthetic/derived data limit real-world interpretation.

The provided common-report template asks for stakeholder and user validation. The supporting materials available for this document did not include a verified list of participants, a usability protocol, a consent record, or the final user-study set of data. As such, no claims can be made that formal user acceptance testing or stakeholder validation has taken place.

In order to finish this evaluation correctly, the team should conduct a structured walkthrough with a few representative suppliers and buyers, define success criteria for tasks ahead of time, record completion time and errors, gather a short usability questionnaire, and analyze qualitative feedback. It is important to tell the participants that the platform is a research prototype and that its predictions can be wrong. Results should identify sample size and recruitment method and should avoid generalizing from the small convenience sample to the whole Sri Lankan textile sector.

6.4 Stakeholder and user validation

The provided common-report template asks for stakeholder and user validation. The supporting materials available for this document did not include a verified list of participants, a usability protocol, a consent record, or the final user-study set of data. As such, no claims can be made that formal user acceptance testing or stakeholder validation has taken place.

In order to finish this evaluation correctly, the team should conduct a structured walkthrough with a few representative suppliers and buyers, define success criteria for tasks ahead of time, record completion time and errors, gather a short usability questionnaire, and analyse qualitative feedback. It is important to tell the participants that the platform is a research prototype and that its predictions can be wrong. Results should identify sample size and recruitment method and should avoid generalising from the small convenience sample to the whole Sri Lankan textile sector.

6.5 Achievement of common objectives

The current state of evidence indicates that we cannot adopt the described goals for the group with full confidence. Both component models and a linked workflow were described in the project description, but some of the systems' claims need more thorough evaluation.

Objective	Current evidence described in project records	Status for common report	Evidence still required
GO1: Define integrated workflow	Four component responsibilities and end-to-end flow documented	Partly achieved / documented	Verify actual workflow in current running build
GO2: Integrate material, sustainability and commercial outputs	Component APIs and seller workflow described	Partly achieved pending end-to-end evidence	API integration test results, request/response examples
GO3: Provide marketplace forecasts	Models and reported metrics documented	Prototype capability reported	Reproducible evaluation, calibration and leakage checks
GO4: Evaluate component outputs	Metrics available for multiple components	Partly achieved	Dataset provenance, splits, per-class metrics and test scripts
GO5: Evaluate end-to-end integration and usability	Integration strategy can be defined; no verified user study supplied	Not established by supplied evidence	Signed test matrix, failure-path results, user study
GO6: Assess commercial and sustainability value responsibly	Limitations and intended use documented	Conceptual assessment completed	Real transaction data, validated LCA assumptions, stakeholder evidence

6.6 Integrated limitations and threats

The main threat to validity is that many results used synthetic/generated data. Such data may be useful for software development, but they have too deterministic relationships and simplified class boundaries which may make prediction easy compared to actual operations. The market may evolve over time, and the relationship between fabric features and commercial outcomes may vary with buyer, district, quantity, or season.

Other threats include target leakage, duplicates across splits, class imbalance, label noise, uncalibrated confidence scores, inconsistent units, missing values, model drift, and user overreliance on numeric outputs. Sustainability indicators are subject to modeling choices about recycling yield, production avoided, transport, energy mix, and end-of-life pathways; these threaten require separate treatment–a high R^2 is no cure. An integration risk occurs if the model expects fields the application does not collect or if fallbacks generate estimates from rules rather than trained models. A prediction endpoint should disclose when a fallback was used. For deployment, the team should add data-quality monitoring, confidence calibration, drift checks, model approval gates, and a documented procedure for withdrawing a harmful or degraded model.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

TexCycle AI's proposed value to the market is found in its ability to reduce the information friction in the exchange of textile waste. This includes suppliers structuring and presenting their lots, buyers finding suitable feedstock, and the system providing decision support on the material type, sustainability indicators, reference price, and indicative marketplace behaviour. This suite of benefits is likely to find application in the contexts described, where information asymmetry reduces informal exchange and limits opportunities for recycling.

The intended beneficiaries of this technology are textile manufacturers that seek alternatives to landfilling, upcycling companies that need appropriate supplies, recyclers that want to identify specific materials, designers and small-scale producers that seek to acquire textile waste in smaller quantities, and administrators that want to improve marketplace oversight. These benefits are hypothesized, as the report provides limited evidence on whether these technologies were actually adopted and used by the target audience. As such, the report is distinguishing between the intended value and the actual impact

7.2 Market alternatives and differentiation

Potential alternatives include direct supplier-buyer relationships, broker networks, general classified marketplaces, material recovery networks, and enterprise resource-planning systems with internal waste records. Such variations may already address some aspects of the problem. The project's novelty is not the generic declaration of a marketplace novelty: it is the proposed combination of textile-specific image analysis, circularity interpretation, price and buyer decision support, marketplace forecasting, and lifecycle traceability in a Sri Lankan context.

The competitive comparison should be drawn on the observable criteria, including textile-specific attributes, buyer discovery, quality/condition information, traceability, environmental transparency, fit within local operations, and ease of use.

The team should not declare that a competitor lacks a feature unless a current and fair comparison has been undertaken.

7.3 Business model and feasibility

A possible implementation at the first stages could be the launch of a simplified listing service and the option to subscribe for verified supplier accounts, later complemented by enhanced analytics, enterprise subscriptions, or a commission-based transaction model. All of these options should be explored to identify the most viable one, but none is likely to be lucrative at early stages or require high expenditures on hosting, moderation, customer support, and other services. The team should calculate the costs of hosting and inference, moderation and account support, and the customer acquisition rates and define customer lifetime value and willingness to pay on the supply and demand sides.

The first launch could be a restricted beta with a reduced category selection and a limited number of supplier accounts and have a goal of gathering demand from buyers. It will be important to observe the number of listings and the ratio of invalid/valid listings, the ratio of enquiries/completed transactions, average matching time, dispute resolution rate, repeat usage rate, and the level of trust from buyers. The environmental impact of the platform should be evaluated only when the supply chain involves physical materials that can be quantified at the destination.

Model option	**Potential value**	**Main cost / risk**	**Validation question**
Free basic listings	Builds supply and demand density	Low monetisation; moderation costs	Will suppliers create accurate, current listings?
Premium supplier analytics	Supports inventory and planning	Need for dependable forecasts	Will businesses pay for improved decisions?
Enterprise subscription	Predictable recurring revenue	Integration and support burden	Can a facility integrate existing

			waste records?
Transaction fee	Revenue aligned with exchange	May discourage off-platform reporting	Can completed transactions be verified fairly?
Verified listing service	Trust and quality differentiation	Inspection and verification cost	Which verification level is valuable to buyers?

Table 5 Potential commercial models and validation questions

7.4 Adoption intellectual property and compliance

Adoption barriers consist of low digital readiness, textile name inconsistency, inventory disclosure hesitation, logistics restrictions, algorithmic estimate mistrust, and a lack of validated transaction history. The platform should offer a transparent explanation of the suggested actions and corrections and refrain from implying that algorithmic suggestions are obligatory.

Intellectual-property considerations consist of ensuring the licenses of code, datasets, images, pretrained weights, and external packages. The tool should collect personal and commercial data only when necessary, utilise restricted access and roles, and store the information on a need-to-know basis. The team should review relevant Sri Lankan data-protection and electronic-transaction regulations with legal authorities before launch, which this report cannot offer.

I think that responsible use of AI involves documenting provenance, limitations, evaluation conditions, and use cases for the data it uses. The model's outputs should be monitored for drift and disproportionate error rates. Wherever possible, the user should be able to identify a generated estimate, its unit of measurement, and have recourse to challenge or amend the input responsibility. A person should always be accountable for commercial decisions and environmental claims. [17], [18].

7.5 Sustainable Development Goal contribution

The service furthers SDG 12, to which it is aligned, by facilitating information exchange for reuse and recovery, and potentially contributes to SDG 8, by improving access to material opportunities for small businesses. Impact assessment for a credible sustainability reporting would benefit from measuring kilograms reused or recycled, the share of material listed that reaches a reported end use, and the avoidance of disposal where a counterfactual can be defended, plus supplier and buyer participation, and economic impact such as value recovered or repeat transactions.

Negative impacts of a marketplace should also be considered: increased transport distances, encouragement of low-value movements of materials, creation of false environmental claims, or exclusion of smaller players if paid listing becomes necessary, among others. These can be mitigated by local matching, disclosure of environmental assumptions, low bandwidth demand, fair listing policies, and careful communication of uncertainties. Impact reporting should specify system boundaries and avoid counting the same material twice

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

This report details one common approach to the development of Tex Cycle AI as a multi-faceted AI-assisted marketplace solution to problems of textile-waste identification, sustainability interpretation, commercial exchange, and marketplace planning. The central challenge is the distribution of meaningful information and effective buyer-seller coordination rather than the singular prediction model. The four components answer different questions and become more interesting when their outputs are related through consistent data contracts and a traceable workflow.

The project documentation reports encouraging prototype metrics across fabric classification, sustainability regression, price prediction, buyer matching, demand classification, listing-success classification, sales-time regression, and market-trend classification. These results illustrate development progress, but the marketplace dataset is described as synthetic, and the component experiments use different tasks and targets. Therefore, it does not establish production performance or real-world environmental benefit. The most defensible conclusion is that the integrated approach is technically plausible, and warrants controlled real-data validation.

8.2 Principal integrated contribution

The main idea of the concept is that it unifies the four areas in a single process of textile waste processing: (1) fabric detection and condition signals; (2) circular and environmental indicators; (3) reference prices, buyers' recommendations, and traceability; and (4) marketplace demand, success, time, and trend estimations. Therefore, the architecture offers a chance to get from a material record to the corresponding list of items with a certain degree of possibility, on the one hand, and the actual marketplace outcomes, on the other hand.

The common report also proposes a common perspective on evaluation and governance. It underlines the importance of consistent unit and identifiers, reliance on evidence, justification of claims, tracing model versions, and addressing failures

on the API side, realistic commercial validation, and assumptions on sustainable practices. These are all-important aspects that have to be considered for creating an integrated system that can be trusted beyond one successful experiment.

8.3 Future directions

The prioritization of future work should be dictated by the evidence gap that the activity closes. The first priority is acquiring a representative, consented, and quality-controlled real dataset including material attributes, listing events, offers, and confirmed transaction outcomes. The second is making the model evaluation reproducible with clear train/validation/test splits and temporal holdouts where appropriate, per-class metrics, calibration, and baseline comparisons. The third is end-to-end testing and a small stakeholder pilot with a documented protocol.

Subsequent priorities would be improving image robustness against lighting and backgrounds, collecting real buyer feedback to evaluate recommendations, measuring listing outcome without target leakage, and model calibration, as well as validation of environmental factors and life-cycle boundaries, data-quality and drift monitoring, and an evaluation of operational costs. Production deployment would require a security and privacy review, tested backup and recovery, clear user support processes, and explicit acceptance criteria. The project should not move from testing on synthetic data to claims about market or environmental impact without this additional evidence.

• Priority 1 is that of the dataset validity - documenting its sources, permissions, labels, units, missingness and provenance.

• Priority 2 is obtaining a reproducible evaluation of the performance: versioning datasets and code and using holdout splits and suitable baselines.

• Priority 3 is verifying the integration - checking success and failure paths across all four components.

• Priority 4 is a stakeholder-based pilot - evaluating the solution’s usability, trust, commercial usefulness and adoption barriers.

• Priority 5 is responsible deployment - implementing monitoring, security, privacy, model governance and impact reporting

REFERENCES

[Guidance - delete before submission: Include only sources cited in this common report. Use one referencing style consistently; IEEE is recommended for IT4010 unless the department instructs otherwise. Single-space each entry and leave one blank line between entries.]

[[1] Replace with the first complete reference entry.]

APPENDICES

The following is an example contract of implementation to guide and verify. It is not necessarily an exact copy of the currently implemented API. Replace endpoint names, field names, authentication details and response examples with a real implementation before submitting this file.

Appendix A Integrated interface specification

Endpoint / contract	Purpose	Example fields	Verification note
POST /api/materials (illustrative)	Create or update a material lot	material_id, fabric_type, weight_kg, condition, district	Confirm actual route and required fields
POST /api/sustainability/predict (documented project route)	Return sustainability outputs	fabric_type, weight_kg, health_score, repairability, condition, district, province, industry_type	Confirm schema, units and error payload
Marketplace prediction endpoint (confirm route)	Return marketplace forecasts	demand_level, listing_success_probability, estimated_days_to_sell, market_trend	Confirm actual route and feature names
Price prediction endpoint (confirm route)	Return reference price	predicted_price_per_kg, currency, model_version	Confirm whether prediction is per kg or total
Traceability event creation (confirm route)	Record lifecycle change	event_id, listing_id, event_type, timestamp, actor_id	Ensure only authorised state transitions

Appendix B Team contribution declaration

The JSON below is an illustrative shape only. The final report must use the exact schema captured from the running service or API specification.

```
{
  "material_id":"MAT-EXAMPLE-001",
  "fabric_type":"denim",
  "weight_kg":50,
  "condition":"good",
  "district":"confirm-supported-value",
  "province":"confirm-supported-value"
}
```

An illustrative successful response should identify the requested material or request ID, the prediction fields, their units, the model version, and the timestamp. An error response should identify a stable error code and whether retry is appropriate, and should not expose stack traces, credentials, internal file paths, or personally identifiable information.

Appendix C — Integrated Test Matrix

Test ID	Scenario	Expected result	Evidence to attach	Status
INT-01	Valid textile record submitted	Record is accepted and receives a stable identifier	Request/response capture and database record	To be executed/verified
INT-02	Required field missing	Validation error returned; no malformed record created	API test output	To be executed/verified
INT-03	Unsupported fabric class	Clear validation or controlled unknown-class behaviour	API/UI capture	To be executed/verified
INT-04	ML service timeout	User sees unavailable/retry state; no fabricated prediction	Server log and UI capture	To be executed/verified
INT-05	Model artifact unavailable	Service fails safely and logs diagnostic identifier	Service log/test	To be executed/verified

INT-06	Unauthorised listing edit	Request denied; data unchanged	Role test output	To be executed/verified
INT-07	Duplicate create request	No unintended duplicate transaction or event	Database/query evidence	To be executed/verified
INT-08	Invalid probability output	Response rejected or flagged; UI does not present invalid value	Unit/API test	To be executed/verified
INT-09	Completed offer/transaction	Correct state transition and event history	Record history capture	To be executed/verified
INT-10	Prediction history retrieval	Only authorised user's records are returned	API test output	To be executed/verified
INT-11	Regression after model update	Contract and expected behaviour tests still pass	CI/test report	To be executed/verified
INT-12	Mobile/narrow viewport	Core actions remain readable and usable	Actual device/browser screenshots	To be executed/verified

Table 6 Integrated test matrix

Appendix D — Model and Data Provenance Checklist

• Record dataset name, source, collection dates, licensing/consent, owner

• Record whether the dataset is real, synthetic, rule generated, augmented or mixed.

• Document target definitions, units, class mapping, label generation process, known label noise

• Record train/validation/test split method, random seed, duplicate handling, leakage checks

• Fit preprocessing on training data only, and persist compatible preprocessing for each model.

• Record model family, hyperparameters, training code revision, library versions, model artefact version, evaluation report.

• Report class balance, per class precision/recall/F1, confusion matrix, regression residual analysis where appropriate

• Note the difference between model level evaluation and evaluation of a single prediction confidence or score.

• Define triggers for retraining, approval criteria, rollback procedure and post deployment monitoring.

• Review privacy, bias, security and unintended use risks prior to release.

Appendix E — Team Contribution Declaration

The team should jointly review this matrix, correct it according to the actual division of work, and sign the final version. The labels below describe the component responsibilities from the project documentation and are not meant to replace a signed individual contribution record.

Contribution area	Member 01	Member 02	Member 03	Member 04	Shared
Fabric classification / condition	Lead				Integration
Sustainability models / indicators		Lead			Integration
Pricing / buyer recommendation / traceability			Lead		Integration
Marketplace predictions / analytics				Lead	Integration
Common architecture and data contracts					All members
End-to-end testing					All members
Report editing and final verification					All members

Table 7 Team contribution declaration

Appendix F — Screenshot And Evidence Plan

The team should walk through this matrix together, check it against the actual division of work and sign off the final version. The labels below represent

component responsibilities from the project documentation and do not replace a signed individual contribution record.

The following captures should be embedded only once they have been extracted from the actual current build. Each screenshot should be clear, cropped to the relevant interface and cited in the main text or evidence appendix. Remove any test accounts, secrets, personal data or irrelevant browser information in advance of inclusion.

- F.1 Seller dashboard indicating the actual current dashboard and signed-in role.
- F.2 Upload Textile Waste form indicating the real field labels and validation behaviour.
- F.3 Fabric analysis result against a documented sample image.
- F.4 Sustainability Analysis result including units and explanatory labels.
- F.5 Reference price and buyer recommendation screen against a real test record.
- F.6 Marketplace demand, listing-success, sales-time and trend outputs against a documented test listing.
- F.7 Listing details and buyer-facing discovery/search page.
- F.8 API test output against a successful request and a controlled invalid request.
- F.9 Traceability or prediction history screen if implemented.
- F.10 Test runner output and selected integration test evidence

Appendix G — Risk Register

Risk	Likelihood / impact (initial qualitative view)	Mitigation	Owner / review
Synthetic data does not represent actual market	High / High	Collect representative real data; re-evaluate before external claims	All members; each dataset revision
Target leakage in marketplace models	Medium / High	Feature audit; temporal split; remove post-outcome fields	Member 04; before each evaluation
Incorrect environmental interpretation	Medium / High	Versioned factors, units, boundaries, uncertainty, expert review	Member 02; each factor/model update
API contract drift	Medium / High	Schema versioning, contract tests, integration gates	Component owners; each release
Prediction overreliance	Medium / High	Plain-language caveats, human review, correction path	All members; UI review
Security/privacy incident	Medium / High	Least privilege, secrets management, logging controls, incident plan	System owner; before deployment
Poor marketplace adoption	Medium / Medium	Pilot, stakeholder feedback, simplified onboarding	Product/team lead; each pilot
Model drift after launch	High / High	Monitor data and outcomes; define retraining/rollback policy	Relevant model owner; monthly or threshold-based

Table 8 Initial integrated risk register

Appendix H — Recommended Release Checklist

• Confirm project title, degree title, department, members names/IDs, and supervisor.

• Replace any identifying details using the approved university group record.

• Check all in-text citations appear in the reference list and every reference is cited.

• Replace illustrative examples of APIs with actual endpoint/schema evidence.

• Insert application and testing screenshots where indicated in Appendix F.

• Execute and report on the outcome of the integration tests; replace 'To be executed/verified' with what is observed.

• Update contents, figure/table lists, cross-references, and page numbers in Microsoft Word.

• Check all figures/tables are numbered according to the chapter number and that each is mentioned in the text.

• Review any requirements around the disclosure of use of AI against the module policy and complete as required.

• Check spellings, units of measure, currency symbols, metric definitions, confidentiality, and all claims with the team.

• Export the PDF version of the final Word file and check page breaks, tables, captions, images, and page numbers.

• Obtain supervisor feedback and make all required corrections before submission

Appendix I — Evidence Limitations And Interpretation

This report deliberately makes the distinction between documented prototype metrics, conceptual architecture, recommended tests, and claims that need further evidence. The figures in Chapter 6 are taken verbatim from project documentation and should be cross-referenced with the final component evaluation scripts and individual theses. A metric should not be presented as one's independent reproduction unless the underlying experiment has been redone and documented.

The architecture figures are original explanatory diagrams created for this report. They are not screenshots of a deployed application. The test matrix is a plan until executed; a blank or planned test is not a passing test. The team should maintain this distinction in the final submission in order to preserve academic integrity and avoid overstating the project's maturity

Appendix J — Shared Data Dictionary

This data dictionary provides a suggested shared vocabulary for integration and should be reconciled with the current database schema / API implementation before the report is submitted. The presence of fields below does not imply that they are currently implemented.

Field	Meaning	Type / unit	Owner / source	Validation or interpretation
material_id	Stable identity of the physical material lot	String / identifier	Shared application	Required; must not be reused for another lot
listing_id	Identity of a published offer for a material lot	String / identifier	Marketplace backend	A material may have one or more listing records only if the business rules allow it

fabric_type	Predicted or confirmed fabric category	Controlled category	Fabric Intelligence / seller correction	Store model suggestion separately from seller-confirmed value
fabric_confidence	Model confidence for predicted category	Number [0,1]	Fabric model	Not equivalent to accuracy on the full test set
condition	Condition selected or assessed for the lot	Controlled category	Seller / fabric workflow	Allowed values must be agreed across services
defect_probability	Image-level defect-class probability, if exposed	Number [0,1]	Fabric model	Do not describe as defect area or segmentation mask
health_score	Condition-related score	Documented score scale required	Fabric / sustainability interface	Specify scale and whether higher is better
repairability	Repairability category or score	Controlled category / scale	Fabric / sustainability interface	Version the category vocabulary
remaining_lifespan	Estimated remaining useful life	Numeric; unit required	Fabric workflow if available	State assumptions; avoid false

				precision
weight_kg	Material lot mass	Kilograms	Seller	Must be positive and finite
district	District associated with the material/listing	Controlled location value	Seller / marketplace	Validate spelling and supported values
province	Province associated with the material/listing	Controlled location value	Seller / marketplace	Should agree with district mapping where available
circularity_score	Model-estimated circularity indicator	Score scale required	Sustainability model	Not a measured environmental outcome by itself
recyclability_score	Model-estimated recyclability indicator	Score scale required	Sustainability model	Depends on material and processing context
co2_saved_kg	Estimated CO_2-related saving	Kilograms; factor basis required	Sustainability model	Requires baseline, boundary, and factor version
water_saved_liters	Estimated water saving	Litres; factor basis	Sustainability model	Requires baseline,

		required		boundary, and factor version
reference_price_per_kg	Suggested price per kilogram	LKR/kg if implemented that way	Commercial Intelligence	Clearly distinguish reference price from seller asking price
buyer_category	Suggested buyer segment	Controlled category	Commercial Intelligence	Recommendation is not a transaction guarantee
demand_level	Predicted demand category	Low / Medium / High, if used	Marketplace Analytics	Record model version and prediction timestamp
listing_success_probability	Estimated probability of sold outcome	Number [0,1]	Marketplace Analytics	Calibrated probability only if calibration is evaluated
estimated_days_to_sell	Predicted days to sell	Days	Marketplace Analytics	Do not show negative values; state prediction horizon
market_trend	Predicted trend category	Growing / Stable / Declining	Marketplace Analytics	Class confidence is not model validation

				accuracy
model_version	Version of the model used	String	ML service	Store alongside relevant predictions
created_at	Creation time of record/event	ISO 8601 timestamp	Application backend	Use consistent timezone policy
event_type	Lifecycle state change	Controlled event type	Traceability layer	Validate allowed state transitions
prediction_source	Model or fallback/rule source	Controlled value	ML/API layer	Expose fallback use for audit and interpretation

The shared data dictionary is a governance artefact which should be under review when a model changes its anticipated characteristics, when a user interface changes any field meaning or if an environmental indicator changes its unit or factor basis.

Appendix K — Component Model Card Summary

These summaries consolidate the model roles reported in the project documentation. They are not substitutes for the detailed model cards in each individual thesis. Any missing dataset or training metadata must be completed from the actual training scripts and evaluation outputs.

Component/model	**Task and reported method**	**Reported evidence**	**Known limitation / required addition**

Fabric classification	Multiclass image classification; EfficientNet-B3; 16 classes documented	Accuracy 89.68%	Add dataset size, per-class precision/recall/F1, split method and failure examples
Defect classification	Binary whole-image classification; MobileNetV2 pathway	Accuracy 90.87%	Do not claim segmentation; add class balance, threshold policy and error analysis
Sustainability regressors	Six XGBoost regressors for circularity, recyclability, carbon/water reductions and savings	Six MAE/RMSE/R^2 triplets in Chapter 6	Document synthetic/derived labels, units, baseline assumptions and independent factor validation
Sustainability grade classifier	Separate classifier and preprocessing pipeline	Model is documented; detailed classifier metrics not included in common source excerpt	Add confusion matrix, class-wise metrics, label definition and calibration
AI price prediction	XGBoost regressor	MAE 12.24; RMSE 15.40; R^2 0.9590	Confirm currency/target unit, target range, test size, baseline

			and real-data validity
Buyer recommendation	Random Forest + XGBoost documented for ranked buyer categories	Matching accuracy 83.56%	Define relevance/ground truth, top-k metrics and user acceptance evidence
Demand prediction	Trained classifier; low/medium/high categories described	Accuracy 99.54%	Investigate synthetic-label construction and potential leakage
Listing success	XGBoost classifier; sold/unsold target	Accuracy 83.65%	Report precision/recall/F1, class balance, temporal split and calibration
Sales-time prediction	XGBoost regressor; days-to-sell target	R^2 0.7610	Add MAE/RMSE, units, censoring treatment and time-based validation
Market trend	Trained model with label encoder; Growing/Stable/Declining classes	Accuracy 98.72%	Add per-class metrics, class balance, horizon definition and time-aware evaluation

Table 9 Component model card summary

Recommended model card fields: out-of-scope use, intended use, training data provenance, data preprocessing, model architecture, version, evaluation split, metrics, known limitations, subgroup/error analysis, runtime dependencies, owner, deployment date, monitoring thresholds, rollback process.

APPENDIX L — EXTENDED TEST CASES AND ACCEPTANCE CRITERIA

ID	Test / setup	Procedure	Expected acceptance criterion	Evidence
API-01	Valid required fields	Submit a valid material record	Success response; stable identifier; stored values match request	Automated test output + database check
API-02	Invalid numeric quantity	Submit zero, negative, NaN or non-numeric weight	Request rejected with validation error; no invalid record stored	Test output
API-03	Unknown categorical value	Submit fabric/condition value outside controlled vocabulary	Clear rejection or explicit unknown handling; no silent mis-encoding	Test output
API-04	Service	Stop or block	UI indicates	UI capture +

	unavailable	the ML service in a controlled test	estimate unavailable; listing remains consistent	server log
API-05	Model version change	Run the same fixed test set against approved model version	Output schema remains compatible; regression metrics reviewed	Release note + test report
SEC-01	Unauthenticated protected request	Call protected route without valid credentials	Request denied; no protected data disclosed	HTTP response capture
SEC-02	Wrong-role update	Use buyer role to change a seller-only field	Permission denied; database unchanged	Role-based test result
SEC-03	Sensitive error details	Trigger controlled server error	Client response excludes stack trace, secrets and local paths	Response capture
FLOW-01	Complete seller journey	Create record, review outputs and publish listing	All required stages complete with same listing/material IDs	Screen recording or screenshots

FLOW-02	Partial component failure	Make one prediction service fail while others respond	Completed outputs retained; failed stage clearly identified and retryable where safe	Integration test
FLOW-03	Duplicate event submission	Submit same event/request twice	No unintended duplicate transaction or state transition	Database/event query
FLOW-04	History access control	Request another user's prediction/listing history	Access denied or records filtered correctly	API test
ML-01	Train/test leakage audit	Inspect feature list and split pipeline	Post-outcome variables excluded from pre-outcome predictions	Code review checklist
ML-02	Class imbalance check	Report target counts and per-class metrics	Accuracy accompanied by class-wise evidence	Evaluation report
ML-03	Temporal validation	Train on earlier records and test on later records where timestamps	No future observations leak into training features	Split definition and metrics

		support it		
UX-01	Prediction explanation	Ask a user to interpret a forecast label and probability	User can distinguish estimate from guarantee; confusion recorded	Usability notes
OPS-01	Backup restoration	Restore a backup into a test environment	Data and configuration restored to documented recovery point	Restoration record
OPS-02	Load behaviour	Run a stated concurrent workload	Latency and error-rate targets met or limitations documented	Load-test output

Table 10 Extended integration and acceptance test cases

Appendix M — Sustainability Impact Measurement Plan

A marketplace can enable circularity, but the software's predicted indicator is not necessarily an indicator of environmental benefit. The following measurement plan differentiates between operative activity and impact outcomes and determines what evidence is needed to make a claim.

Indicator	Operational definition to agree	Evidence source	Reporting caution
Material listed (kg)	Sum of distinct valid material quantities published during period	Listing database and duplicate checks	Do not count drafts or duplicate listings
Material sold (kg)	Quantity in completed, confirmed transactions	Transaction record and seller/buyer confirmation	An offer or predicted success is not a sale
Material reused/recycled (kg)	Quantity with verified downstream destination	Delivery/receipt evidence and buyer declaration	Sale alone does not prove actual reuse or recycling
Time to match	Time between valid listing publication and first qualified buyer engagement	Event timestamps	Define qualified engagement and exclude test traffic
Transaction completion rate	Completed transactions divided by accepted offers, with denominator stated	Offer/transaction state history	Explain cancelled, disputed and partially completed cases
Estimated avoided disposal	Verified quantity multiplied by an agreed counterfactual factor	Material destination plus documented factor source	State boundary, factor version and uncertainty
Estimated CO_2-equivalent effect	Difference between defined baseline and alternative	LCA model, factors and process data	Do not present as measured emissions without measurements

	pathway		
Water-related effect	Difference between defined baseline and alternative pathway	LCA model, factor source and process data	Avoid double counting and clarify units
Economic value recovered	Completed transaction value net of defined fees/returns	Transaction records	Do not substitute asking price or predicted price
Participation of small enterprises	Number/share of active verified small suppliers and buyers	Account metadata and agreed definition	Protect business privacy and avoid unsupported classifications

Table 11 Sustainability impact measurement plan

The team should record the reference scenario, system boundary, allocation rules, period of time, data's completeness, and uncertainty before finalizing the environmental impact total. ISO 14040 and ISO 14044 present general life-cycle assessment framework and requirements [4], [5].

Appendix N — Glossary

Term	Definition used in this report
AI-generated estimate	**A value produced by a model or rule-based method; it is not automatically a verified fact.**
Buyer recommendation	**A ranked suggestion of buyer categories or potential purchasers based on available features.**
Circular economy	**An approach that seeks to retain the value of products and materials through reuse, repair, remanufacture and recycling.**
Data leakage	**Use of information during training or evaluation that would not legitimately be available at prediction time,**

	inflating reported performance.
Defect classification	**Whole-image classification of whether a defect is present; it is different from pixel-level defect segmentation.**
Feature	**An input variable used by a model to produce a prediction.**
Ground truth	**A reference label or observed outcome used to train or evaluate a model.**
MAE	**Mean absolute error; average absolute difference between predictions and reference values.**
Model drift	**A change in data or relationships over time that can reduce model performance.**
Model version	**An identifier that links a prediction to the model and compatible preprocessing used.**
R^2	**Coefficient of determination; a regression metric comparing model predictions with target variation under the evaluation setup.**
RMSE	**Root mean squared error; square root of the mean squared prediction error, expressed in the target's units.**
Synthetic dataset	**Artificially generated records used for development or evaluation;**

	representativeness depends on how records and labels were generated.
Traceability event	**A timestamped record of a material, listing or transaction lifecycle change.**
Temporal validation	**Evaluation in which later records are held out from training to better simulate prediction into the future.**

Table 12 Glossary

TEMPLATE CONVENTIONS

[Guidance - delete before submission: Delete this entire section before final submission. It is placed at the end so that the official title page remains the first page while you work.]

- Use the built-in Title, Heading 1, Heading 2, Heading 3, Normal, Caption, and Guidance styles. Do not imitate headings using manual font changes.
- Replace and delete all grey italic guidance text and all square-bracket placeholders before submission.
- Use Times New Roman 12 point for body text and 1.5 line spacing. Abstracts, tables, captions, long quotations, and reference entries may use single spacing.
- Retain A4 page size and the prescribed margins: 40 mm left and bottom; 25 mm top and right.
- Start each chapter on a new page. Use lowercase Roman page numbers for preliminary pages and Arabic page numbers for the main body and appendices.
- Create captions using Word's Insert Caption command. Number figures and tables by chapter and refer to each one by number in the text.
- Before submission, press Ctrl+A and F9 in Microsoft Word to update the table of contents, lists, captions, cross-references, and page numbers.
- Use clear academic language, proofread the final document, disclose AI assistance, and verify every citation, quotation, result, and claim.
- Keep the common report focused on the complete solution, component relationships, integration, overall evaluation, and shared value. Do not paste four individual theses into it.